

Practical 4: Activity 1 & 2 Complete Evidence of Response

Activity 1 — Policy-to-path analysis

Case / Policy	Boolean Rule / Code	Best Construct	Justification
a) Credits == 4	credits == 4 True: 'Core Course' False: No output	Simple if	Only a single specific path requires action. No alternative action exists for other values.
b) Mobile Digits	mobileDigits == 10 True: 'Approved' False: 'Rejected'	Standard if-else	Two mutually exclusive outcomes exist; one block must execute based on validation.
c) Student ID Match	idMatches && nameMatches && deptMatches True: Verified False: Access Denied	Nested if	Requires checking multiple distinct identity data fields sequentially step-by-step.
d) Parcel Fit Check	supportedSize && withinWeight True: 'Accepted' False: 'Review'	Compound if-else with &&	Two independent physical criteria (dimensions and mass) must simultaneously be true.

Activity 2 — Predict before running

Case	Input Scenario	Branch Path Taken	Exact Expected Output
1	credits = 4	if body executed	Course: PSP1 Core Course Registration step
2	credits = 3	if body skipped entirely	Course: PSP1 Registration step
3	mobileDigits = 10	if true branch executed	Approved
4	mobileDigits = 9	else false branch executed	Rejected
5	internal=20, external=30	pass branch criteria met	Status: Pass
6	weightKg = 25.0	review boundary fallback	Decision: Review